$$\frac{\mathrm{d}\hat{\sigma}_z}{\mathrm{d}t} = \mathrm{i}\big[\hat{H}_0, \hat{\sigma}_z\big] = 0, \tag{5.94a}$$

$$\frac{\mathrm{d}\hat{a}_i^\dagger}{\mathrm{d}t} = \mathrm{i}\big[\hat{H}_0, \hat{a}_i^\dagger\big] = \mathrm{i}\omega_i\big[\hat{a}_i^\dagger\hat{a}_i, \hat{a}_i^\dagger\big] = i\omega_i\hat{a}_i^\dagger, \tag{5.94b}$$

$$\frac{\mathrm{d}\hat{a}_i}{\mathrm{d}t} = \mathrm{i}\big[\hat{H}_0, \hat{a}_i\big] = i\omega_i\big[\hat{a}_i^\dagger\hat{a}_i, \hat{a}_i\big] = -\mathrm{i}\omega_i\hat{a}_i. \tag{5.94c}$$